Problem 1: Use the dimension theorem to find the dimensions of the row space, column space, null space of A, and null space of A^T, where A is a 5×9 matrix with $\text{rank}(A) = 2$.

Solution

Let A be 5×9 with $\text{rank}(A) = 2$.

- $\dim(\text{Col}(A)) = \text{rank}(A) = 2$
- $\dim(\text{Row}(A)) = \text{rank}(A) = 2$
- $\dim(\text{Null}(A)) = 9 - \text{rank}(A) = 7$
- $\dim(\text{Null}(A^T)) = 5 - \text{rank}(A^T) = 5 - \text{rank}(A) = 3$

$$\boxed{\dim(\text{Col}(A)) = 2, \quad \dim(\text{Row}(A)) = 2, \quad \dim(\text{Null}(A)) = 7, \quad \dim(\text{Null}(A^T)) = 3}$$

Problem 2: Use the dimension theorem to find the dimensions of the row space, column space, null space of A, and null space of A^T, where A is a 6×8 matrix with $\text{rank}(A) = 3$.

Solution

Let A be 6×8 with $\text{rank}(A) = 3$.

- $\dim(\text{Col}(A)) = \text{rank}(A) = 3$
- $\dim(\text{Row}(A)) = \text{rank}(A) = 3$
- $\dim(\text{Null}(A)) = 8 - \text{rank}(A) = 5$
- $\dim(\text{Null}(A^T)) = 6 - \text{rank}(A^T) = 6 - \text{rank}(A) = 3$

$$\boxed{\dim(\text{Col}(A)) = 3, \quad \dim(\text{Row}(A)) = 3, \quad \dim(\text{Null}(A)) = 5, \quad \dim(\text{Null}(A^T)) = 3}$$

Problem 3: Find a basis for the null space of $A = \begin{bmatrix} 1 & -1 & 3 \\ 5 & -4 & -4 \\ 7 & -6 & 2 \end{bmatrix}$. Comment on the nullity of A.

Solution

Row reduce A:

$$\begin{bmatrix} 1 & -1 & 3 \\ 5 & -4 & -4 \\ 7 & -6 & 2 \end{bmatrix} \xrightarrow{R_2 \leftarrow R_2 - 5R_1,\ R_3 \leftarrow R_3 - 7R_1} \begin{bmatrix} 1 & -1 & 3 \\ 0 & 1 & -19 \\ 0 & 1 & -19 \end{bmatrix}$$

$$\xrightarrow{R_3 \leftarrow R_3 - R_2} \begin{bmatrix} 1 & -1 & 3 \\ 0 & 1 & -19 \\ 0 & 0 & 0 \end{bmatrix}$$

Now solve $A\mathbf{x} = \mathbf{0}$:

$x_2 - 19x_3 = 0 \Rightarrow x_2 = 19x_3, \quad x_1 - x_2 + 3x_3 = 0 \Rightarrow x_1 = x_2 - 3x_3 = 19x_3 - 3x_3 = 16x_3.$

Let $x_3 = t \in \mathbb{R}$, then:

$$\mathbf{x} = t \begin{bmatrix} 16 \\ 19 \\ 1 \end{bmatrix}.$$

$$\text{Null}(A) \text{ basis: } \left\{ \begin{bmatrix} 16 \\ 19 \\ 1 \end{bmatrix} \right\}, \quad \text{nullity}(A) = 1$$

Problem 4: Find a basis for the null space of $A = \begin{bmatrix} 2 & -3 & 1 \\ 4 & -5 & -2 \\ 6 & -9 & -1 \end{bmatrix}$. Comment on the nullity of A.

Solution

Same as problem 3.

Problem 5: Find a basis for the null space of $A = \begin{bmatrix} 1 & 2 & -1 & 3 \\ 2 & 4 & -2 & 6 \\ 3 & 6 & -3 & 9 \end{bmatrix}$. Comment on the rank of A.

Solution

Same as problem 3. Rank will be number of non-zero rows in echelon form.

Problem 6: Find a basis for the null space of $A = \begin{bmatrix} 1 & 0 & 2 & -1 \\ 0 & 1 & 3 & 2 \\ 0 & 3 & 9 & 6 \end{bmatrix}$. Comment on the rank of A.

Solution

Same as problem 3. Rank will be number of non-zero rows in echelon form.

Problem 7: Use the dimension theorem to find the dimension of the row space, column space, and null space of A^T, where A is a 9×5 matrix with $\text{rank}(A) = 2$.

Solution

We begin with:

$$\text{rank}(A) = \text{rank}(A^T) = 2$$

A^T is 5×9, so by the dimension theorem:

$$\dim(\text{Row}(A^T)) = \text{rank}(A^T) = 2$$

$$\dim(\text{Col}(A^T)) = \text{rank}(A^T) = 2$$

$$\dim(\text{Null}(A^T)) = 9 - \text{rank}(A^T) = 7$$

$$\boxed{\dim(\text{Row}(A^T)) = 2, \quad \dim(\text{Col}(A^T)) = 2, \quad \dim(\text{Null}(A^T)) = 7}$$

Problem 8: Use the dimension theorem to find the dimension of the row space, column space, and null space of A^T, where A is a 6×2 matrix with $\text{rank}(A) = 2$.

Solution

We begin with:

$$\text{rank}(A) = \text{rank}(A^T) = 2$$

A^T is 2×6, so by the dimension theorem:

$$\dim(\text{Row}(A^T)) = \text{rank}(A^T) = 2$$

$$\dim(\text{Col}(A^T)) = \text{rank}(A^T) = 2$$

$$\dim(\text{Null}(A^T)) = 6 - \text{rank}(A^T) = 4$$

$$\boxed{\dim(\text{Row}(A^T)) = 2, \quad \dim(\text{Col}(A^T)) = 2, \quad \dim(\text{Null}(A^T)) = 4}$$